MILITARY STANDARD

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

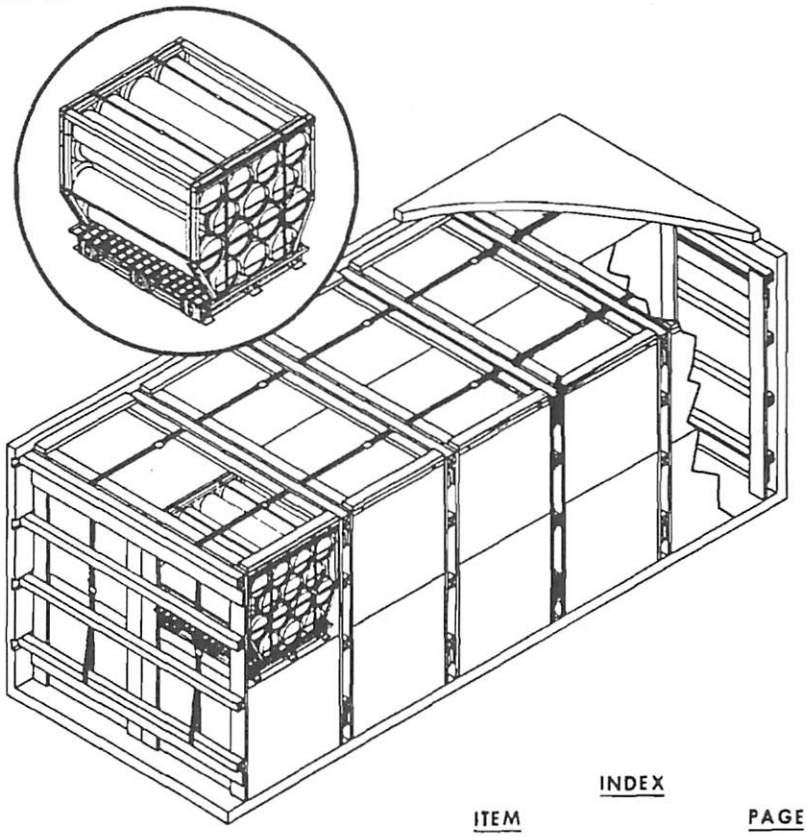
CHARGE, PROPELLING 6"/47 IN CARTRIDGE TANK MK 4 FLEET ISSUE UNIT LOAD

MIL-STD-1386-24
(NAVY)

11 JANUARY 1974

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-54/7
DIMENSIONS	45 1/2 L x 39 3/8 W x 40 3/8 H
CUBE	41.8 CU. FT.
WEIGHT	1317 LBS
DOT HAZARD CLASSIFICATION	EXPLOSIVES B
COAST GUARD CLASSIFICATION	II B



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

INDEX

ITEM	PAGE
GENERAL NOTES	2
TOFC, COFC, CONTAINERSHIP & HIGHWAY	3
DETAILS	4

FSC 8140

AUTHORIZED FOR TRAILER-ON-FLAT-CAR (TOFC), CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS

M.S. Gray NWHL 12/14/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 12/14/73
SIGNATURE ORD SYSCOM BY DIRECTION DATE

APPROVED BY US COAST GUARD

P.J. Donahy
DATE 3 Dec 1973

APPROVED BY BUREAU OF EXPLOSIVES

A.F. Brunnock
DATE 11/28/73

ORIGINATOR A R Casingi
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 20261O-24T00:12Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF 6"/47 PROPELLING CHARGES IN MK 4 CARTRIDGE TANKS. THE UNIT LOAD SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/7.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.
4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT.

CAUTION

WHEN POSITIONING CROSSMEMBERS, MAKE A VISUAL CHECK TO ENSURE THAT CROSSMEMBERS ARE SECURELY IN PLACE.

5. WHEN THE VOID BETWEEN THE CROSSMEMBER AND THE SEPARATOR GATE IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN THE CROSSMEMBER AND GATE AS SHOWN IN DETAIL B.
6. EACH BAY OF UNIT LOADS IS TOMMED DOWN WITH TWO 1 1/4 x .035 STEEL STRAPS. STRAPPING FRAME ASSEMBLY SHALL BE USED ON TOP OF THE TOP UNIT LOADS. THE STRAPS ARE ATTACHED TO THE CROSSMEMBERS AS SHOWN IN DETAIL A. LOCATE SEALS AS SHOWN IN THE BUBBLES OF THE STRAPPING DIAGRAM, DETAIL A. SECURE EACH END OF THE STRAP WITH ONE SEAL WITH TWO PAIR OF NOTCHES PER SEAL OR TWO SEALS, BUTTED TOGETHER WITH TWO PAIR OF CRIMPS PER SEAL.
7. APPLICABLE MATERIAL SPECIFICATIONS:

   DUNNAGE LUMBER - FED. SPEC. MM-L-751

   NAILS - FED. SPEC. FF-N-105, TYPE III, STYLE 10, COMMON BRIGHT

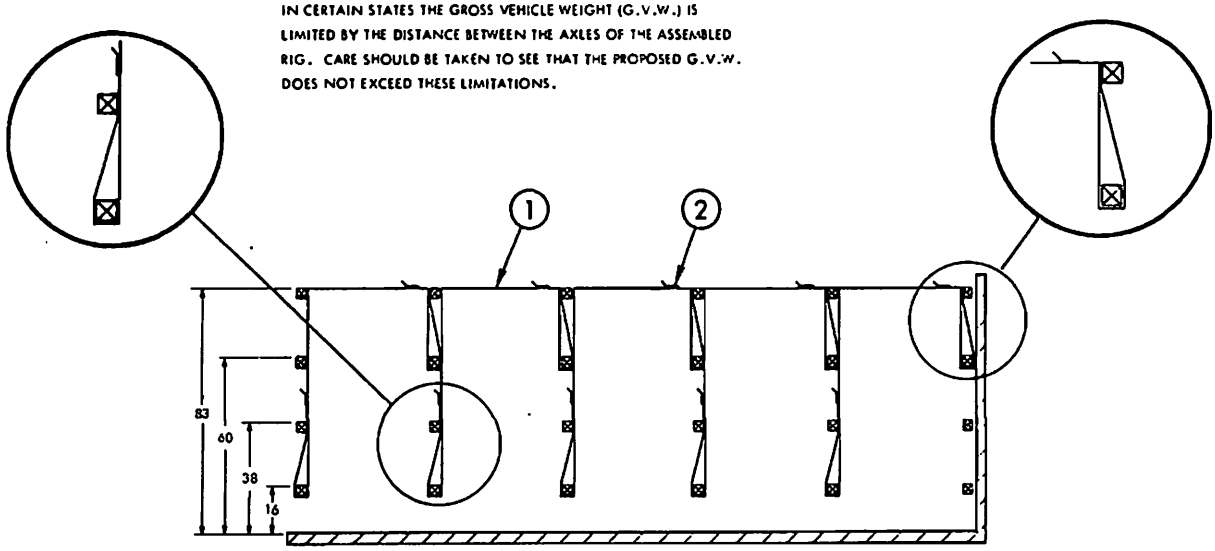
   STRAP - FED. SPEC. QQ-S-781, TYPE I, CLASS A.
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL. II).

TOFC/COFC,CONTAINERSHIP AND HIGHWAY

1. THE CONTAINER LOAD CONSISTS OF 20 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT.
2. THE LOADED MILVAN CONTAINER SHALL BE ON A 20 FT MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES WHEN BEING MOVED IN TOFC SERVICE.
3. WHEN THE LOADED CONTAINER IS IN TOFC OR COFC SERVICE, IT SHALL BE ON FLAT RAILCARS THAT COMPLY WITH BUREAU OF EXPLOSIVES PAMPHLET 6C.
4. THE LOADED MILVAN MAY BE MOVED OVER PUBLIC HIGHWAYS ON A MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR ON SUITABLE COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.

NOTE

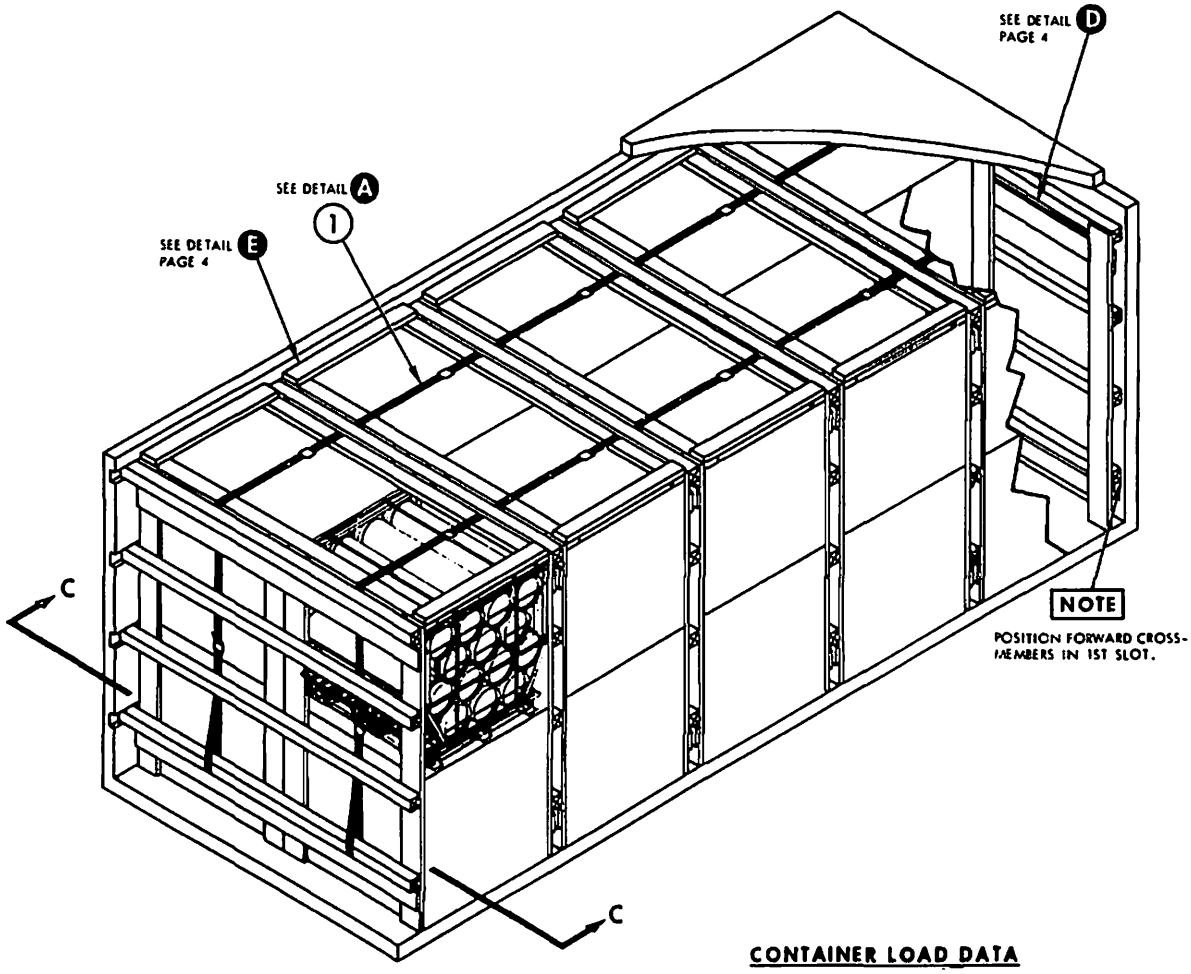
IN CERTAIN STATES THE GROSS VEHICLE WEIGHT (G.V.W.) IS LIMITED BY THE DISTANCE BETWEEN THE AXLES OF THE ASSEMBLED RIG. CARE SHOULD BE TAKEN TO SEE THAT THE PROPOSED G.V.W. DOES NOT EXCEED THESE LIMITATIONS.



DETAIL A

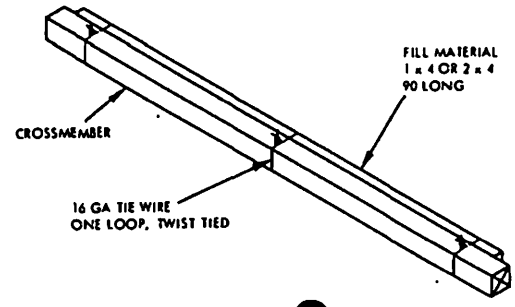
STRAPPING DIAGRAM
INDICATES ATTACHMENT POINTS FOR HOLD-DOWN STRAPPING

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:12Z
Check the source to verify that this is the current version before use.



CONTAINER LOAD DATA

NUMBER OF UNIT LOADS	20
NUMBER OF CROSSMEMBERS REQ'D	24
DUNNAGE WEIGHT	718 POUNDS
CONTAINER WEIGHT	5,785 POUNDS
LOAD WEIGHT	26,340 POUNDS
GROSS WEIGHT	32,843 POUNDS



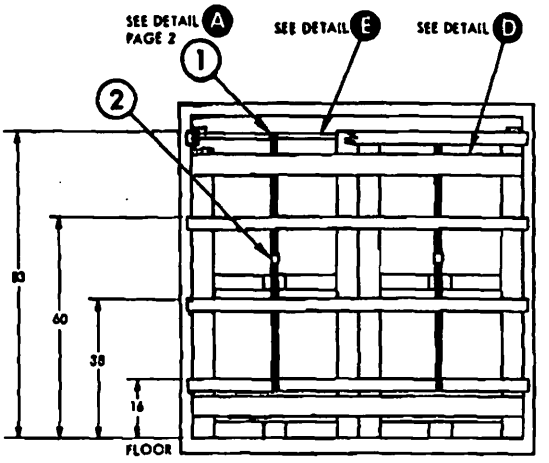
DETAIL B

FILL MATERIAL
SEE NOTE 5

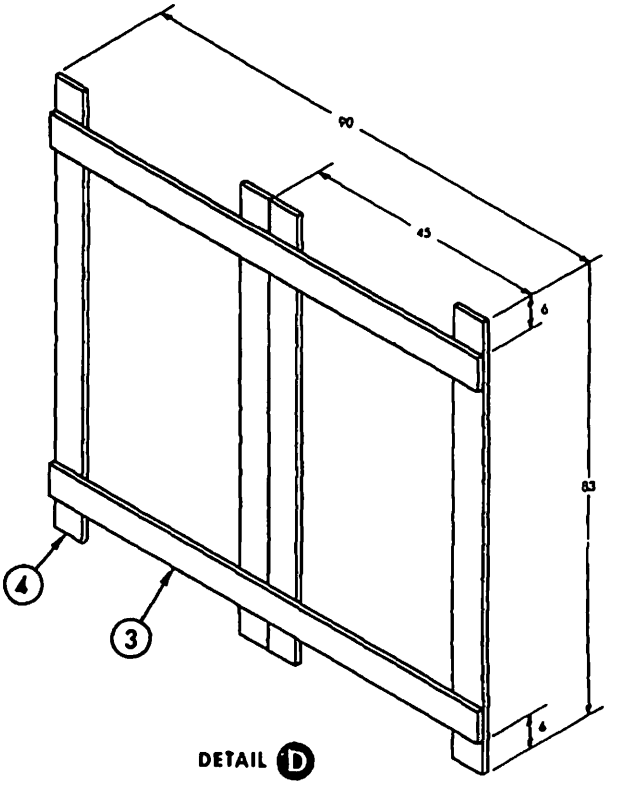
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
6	TIE PIECE	2 x 4 x 39 3/8	10	5	3 PER JOINT	10d
5	HOLD DOWN	2 x 4 x 90	10	SEE 6	-	-
4	VERTICAL	1 x 6 x 83	40	SEE 3	-	-
3	HORIZONTAL	1 x 6 x 90	20	4	3 PER JOINT	8d CLINCHED
2	SEAL	1 1/4	20	-	-	-
1	STRAP	1 1/4 x .035 x 16 FT	10	-	-	-

LIST OF MATERIALS AND NAILING DATA

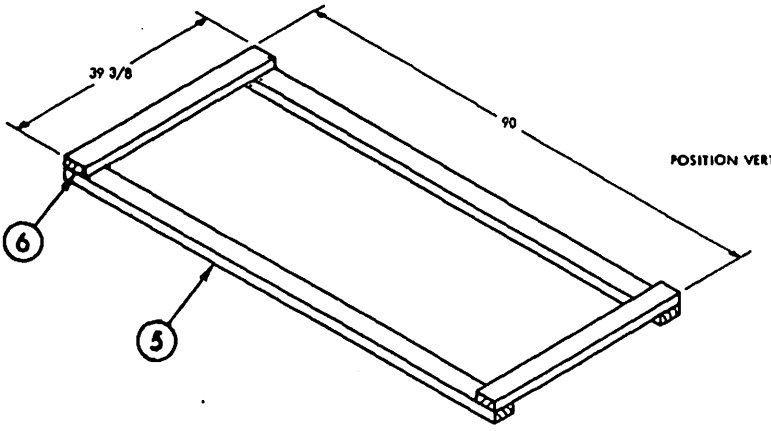
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:12Z
Check the source to verify that this is the current version before use.



SECTION C-C
POSITION VERTICALS AGAINST UNIT LOADS



DETAIL D
SEPARATOR GATE
10 REQUIRED
POSITION VERTICALS, PIECE 4 AGAINST UNIT LOAD



DETAIL E
STRAPPING FRAME
5 REQUIRED

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
PROJECT NO. 8140-N215

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:12Z
Check the source to verify that this is the current version before use.